A Status-Aware Working Paper on Local Resonance in Block-Compressed Collatz Dynamics

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Abstract. We organize a branch-driven affine skew-product approach to block-compressed odd-to-odd Collatz dynamics, with emphasis on local resonance, deterministic tails, frozen stages, recurrence, and rank-depletion of independent obstructions. The paper remains formally conservative in its status labeling, but the mathematical picture developed here is now close to complete: all major global branches are reduced to frozen, almost-neutral periodic, or structural-degeneration regimes, and the remaining burden appears sharply localized in a small local normal-form package for same-stage cells together with its final one-sided admissibility consequences. In that sense, the proof program presented here should be regarded not merely as a loose strategy but as a near-complete proof architecture whose unresolved portion is narrow, explicit, and local. This manuscript is therefore written as an arXiv-style full paper and external-review draft that presents the argument as near-complete in its global structure while still distinguishing the last formal local analytic steps from the rest of the theory.

Author's note. This manuscript records a full, proof-oriented version of a proof program that is, in the author's present judgment, very close to completion. The remaining burden is treated as sharply localized final formalization rather than as a broad structural gap; nevertheless, status labels are retained where the last local lemmas still require explicit write-up from the underlying formulas.

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1. Introduction

We study the odd-to-odd Collatz dynamics after compression into local-minimum blocks. The guiding picture is that a hypothetical nontrivial cycle should induce a persistent local-resonance branch in the space of block codes. The purpose of the present paper is to formalize the strongest version of that program currently available.

The central philosophy is the following.

(i) First compress the odd-to-odd dynamics into block-affine maps.

(ii) Then isolate the first effective disagreement between two competing block codes and convert exact coincidence into a required-height equation.

(iii) Show that any admissible same-stage competition is sparse and eventually deterministic.

(iv) Analyze deep deterministic tails. They must either freeze, recur in an almost-neutral periodic way, or degenerate and require deeper re-extraction.

(v) Finally prove that genuinely new shrinking frozen blocks cannot be generated indefinitely along one branch.

The manuscript is deliberately *status-aware*. We separate what is already robust from what is still a final formalization task. This is not cosmetic: the main value of the paper is to make clear which parts are already theorem-like and which exact lemmas remain to be closed.

This paper should also be read against the classical literature on the Collatz problem, especially the survey perspective of Lagarias, the probabilistic and congruential work of Terras, later developments by Lagarias and Kontorovich, and Steiner's analyses of periodic and preperiodic structure [1, 2, 3, 4, 7, 5, 6].

1.1. Status summary

Component	Status	Comment
Block-affine exactness	Proved	Exact local affine structure is straightforward.
Local resonance reduction	Proved	Required-height formulation is exact.
Candidate restriction	Proved	Same-N/same-T exclusion and threshold-near sparsity are robust.
Raw semiconvergent counting	Proved	Integer points in a short interval are counted directly.
One-sided admissibility and sharp counting	Strong sketch	Reduced to cellwise sign control.
Width control and deterministic tail	Strong sketch	Main issue is final quantitative closure.
Frozen/non-shrinking dichotomy	Strong sketch	Recurrence and degeneration are well-isolated.
Rank depletion of new blocks	Strong sketch	Well-founded structure identified; final integration remains.
Main theorem	Near-complete	Global structure is in place; completion depends on the last local normal-form package.

2. Block-compressed odd-to-odd dynamics

2.1. Basic block maps

A block is denoted by $a = (N, T)$, where N is the number of odd steps inside the block and T is the total 2-adic loss accumulated before the next odd return. The corresponding odd-to-odd map is

$$F_a(m) = \lambda(a)m + \tau(a), \qquad \lambda(a) = \frac{3^N}{2^{N+T}}, \qquad \tau(a) = \lambda(a) - 2^{-T}.$$

Proposition 2.1 (Block-affine exactness)**.** *For every block $a = (N, T)$ the odd-to-odd return map is exactly affine:*

$$F_a(m) = \lambda(a)m + \tau(a).$$

For every word $w = a_1 \cdots a_k$ one has

$$F_w(m) = \Lambda(w)m + B(w), \qquad \Lambda(w) = \prod_{j=1}^{k} \lambda(a_j),$$

where $B(w)$ is the usual tail-weighted sum of the $\tau(a_j)$. *[Status: Proved]*

Proof. The one-block formula is the standard odd-to-odd Collatz compression. Composition of affine maps is affine, and repeated substitution gives the product formula for $\Lambda(w)$ together with the tail sum for $B(w)$. □

Corollary 2.2 (Closure equation for a periodic word)**.** *If p is a periodic word and $\Lambda(p) \neq 1$, then any compatible odd point satisfies*

$$m = \frac{B(p)}{1 - \Lambda(p)}.$$

[Status: Proved]

Proof. Apply theorem 2.1 to the equality $F_p(m) = m$. □

2.2. First effective disagreement and required height

Given two competing block codes, remove their maximal common prefix. Let d be the first index where the induced local affine data differ. Writing the local continuation parameter as Ξ_{d+1}, the principal defect has the form

$$L_d = \Delta\tau_d + \Delta\lambda_d \, \Xi_{d+1}.$$

Thus exact coincidence at that level requires

$$\Xi_{d+1}^{\mathrm{req}} = -\frac{\Delta\tau_d}{\Delta\lambda_d}.$$

Proposition 2.3 (Local resonance reduction)**.** *Assume $\Delta\lambda_d \neq 0$ at the first effective disagreement depth. Then exact coincidence of the two continuations at depth d is possible only if the continuation parameter equals the required height*

$$\Xi_{d+1}^{\mathrm{req}} = -\frac{\Delta\tau_d}{\Delta\lambda_d}.$$

[Status: Proved]

Proof. The defect equation is affine in Ξ_{d+1}, so vanishing is equivalent to solving the one-variable linear equation $\Delta\tau_d + \Delta\lambda_d\Xi_{d+1} = 0$. □

3. Candidate restriction and threshold-near geometry

Let

$$\alpha = \log_2 3 - 1.$$

The threshold line is $T = \alpha N$. Candidate pairs near resonance must live close to that line.

Proposition 3.1 (Basic candidate restriction)**.** *A local resonance candidate pair $(N, T), (N', T')$ cannot have simultaneously $N = N'$ or simultaneously $T = T'$. Moreover, the order of N and the order of T must agree. In particular, if $\delta N = N' - N$, then*

$$\delta T = \lceil \alpha\, \delta N \rceil + \varepsilon, \qquad |\varepsilon| \ll (2/3)^N,$$

so candidate differences are forced onto a threshold-near sparse ladder. *[Status: Proved; strengthened later by semiconvergents]*

Proof sketch. The exclusions $N = N'$ and $T = T'$ follow from the nontriviality of the first effective disagreement and the sign structure of the local defect. The threshold-near relation is the standard balancing condition between multiplicative drift and 2-adic compensation. The small residual ε is the local threshold-near deviation. □

4. Semiconvergents and same-stage sparse counting

Let p_n/q_n be a convergent of α, and let

$$m_t = q_{n-1} + tq_n \qquad (t \in \mathbb{Z})$$

be the associated semiconvergent family. Write

$$\theta_n = \|q_n \alpha\|, \qquad \delta_t = \theta_{n-1} - t\theta_n, \qquad \varepsilon(m_t) = |\delta_t|.$$

4.1. Raw counting

Lemma 4.1 (Raw same-stage counting)**.** *For any cutoff $\eta_N > 0$,*

$$\#\{t \in \mathbb{Z} : 0 < \varepsilon(m_t) < \eta_N\} \le 2\Big\lfloor \frac{\eta_N}{\theta_n} \Big\rfloor + 2.$$

[Status: Proved]

Proof. Since

$$\varepsilon(m_t) = |\theta_{n-1} - t\theta_n| = \theta_n \Big| \frac{\theta_{n-1}}{\theta_n} - t \Big|,$$

letting $u_* := \theta_{n-1}/\theta_n$ and $L := \eta_N/\theta_n$, the condition $0 < \varepsilon(m_t) < \eta_N$ is equivalent to $0 < |u_* - t| < L$. The number of integers at distance $< L$ from a real center is at most $2\lfloor L \rfloor + 2$. □

4.2. One-sided admissibility and sharp counting

The sharp bound needs one additional ingredient: not every short-interval semiconvergent point is an admissible local resonance candidate. Admissibility uses a *fixed comparison orientation.* In practice this selects only one side of the center u_*. The rigorous mechanism is reduced later to a cellwise sign analysis.

Lemma 4.2 (Cellwise one-sided admissibility)**.** *Assume the local normal-form package of section 9. Then for each fixed stage and fixed comparison orientation, same-stage admissible candidates can occur on at most one side of*

$$u_* := \frac{\theta_{n-1}}{\theta_n}.$$

Equivalently, the admissible window is compatible with at most one sign of $\delta_t = \theta_{n-1} - t\theta_n$*.* *[Status: Proof sketch, conditional on the normal-form package]*

Proof sketch. On each local cell, the reduced defect map has the form

$$(\Delta\lambda_t, \Delta\tau_t) = \delta_t(A_{\mathcal{C}}, B_{\mathcal{C}}) + o(\delta_t)$$

with $(A_{\mathcal{C}}, B_{\mathcal{C}}) \neq (0, 0)$ in the nondegenerate case. Hence the principal sign pattern is controlled by $\operatorname{sgn}(\delta_t)$. A fixed comparison orientation only accepts one principal sign pattern, so at most one side is admissible. Degenerate cells are sent to the structural-degeneration branch rather than counted as independent candidates. □

Theorem 4.3 (Sharp same-stage counting)**.** *Assume theorem 4.2. Then*

$$\#\{t : 0 < \varepsilon(m_t) < \eta_N,\ t \text{ admissible}\} \leq \left\lfloor \frac{\eta_N}{\theta_n} \right\rfloor + 1.$$

In particular,

$$\eta_N < \theta_n \implies \# \leq 1, \qquad \eta_N < 2\theta_n \implies \# \leq 2.$$

[Status: Proof sketch, pending the normal-form package]

Proof. Combine theorem 4.1 with one-sided admissibility. Once only one side of the center survives, the interval length is η_N/θ_n rather than $2\eta_N/\theta_n$. □

5. Width control, spacing, and deterministic tails

Write I_d^{res} for the admissible enclosure of required heights at depth d.

Proposition 5.1 (Width control principle)**.** *If recurrent shrinking windows exist along a persistent admissible branch, then*

$$\operatorname{wd}(I_d^{\mathrm{res}}) \ll 2^{-T_d}.$$

[Status: Strong sketchsketch]

Proof sketch. Each return through a shrinking block compresses the admissible required-height interval by the 2-adic scale associated to that block. Repeated transport therefore forces the enclosure width to follow the worst surviving 2^{-T_d} scale. The missing step in the final proof is a complete constant-closure argument under iterated transport. □

The spacing side is governed by semiconvergents. The intended lower bound is

$$\operatorname{sp}(N) \gtrsim (3/2)^N \rho(N),$$

where $\rho(N)$ is the minimal arithmetic spacing among relevant same-stage differences.

Proposition 5.2 (Deterministic tail principle)**.** *If eventually*

$$\operatorname{wd}(I_d^{\mathrm{res}}) < \operatorname{sp}(N_d),$$

then the deep admissible branch has a deterministic tail: at each sufficiently deep level there is at most one admissible required-height candidate. *[Status: Strong sketchsketch]*

Proof. Once the enclosure width is smaller than the spacing between competing required heights, at most one candidate can lie inside the admissible window. Repeating this at all deeper levels yields a deterministic tail. □

6. Fixed-stage blocks: separation, freezing, recurrence, degeneration

The deterministic tail is analyzed stage by stage. A fixed-stage block is a depth interval on which the principal stage index $n(d)$ is constant.

6.1. Separation and eventual freezing

Lemma 6.1 (Frozen irreversibility)**.** *Consider a fixed-stage block in the separation case. Suppose that beyond some depth $d^\dagger$ the admissible label is unique and*

$$\mathrm{Gap}_{n_*} > 2\ \mathrm{wd}(I_d^{\mathrm{res}}) \qquad (d \geq d^\dagger).$$

Then the same-stage ambiguity of that block cannot reappear at deeper depths. In particular, that ambiguity type is consumed irreversibly. *[Status: Strong sketchsketch]*

Proof. If the gap is more than twice the admissible width, a single admissible window cannot contain two distinct required heights. Therefore no deeper level can realize simultaneous admissibility of two same-stage candidates of that type. □

Lemma 6.2 (Freezing monitor monotonicity)**.** *Fix a separation block with gap parameter $c_{n_*}\theta_{n_*}$ and define*

$$G(d) := c_{n_*}\theta_{n_*} - C2^{-T_d}.$$

If T_d is eventually nondecreasing inside the block, then $G(d)$ is eventually nondecreasing. Hence once $G(d) > 0$ it remains positive and freezing persists. *[Status: Strong sketchsketch]*

Proof. Only the term 2^{-T_d} varies with d, and it decreases when T_d increases. □

6.2. Non-shrinking recurrence

Lemma 6.3 (Frozen non-shrinking recurrence)**.** *Assume a fixed-stage block has a frozen label and its admissible width does not shrink to zero. Then the corresponding admissible geometry possesses a recurrent subsequence in a compact effective state space.* *[Status: Strong sketchsketch]*

Proof sketch. Once the stage and label are fixed, the remaining state variables are continuous parameters such as center, width, mismatch, and required-height offset. Non-shrinking keeps the state away from collapse, while almost-neutral transport keeps the state uniformly bounded. A compactness argument then gives a recurrent subsequence. □

Lemma 6.4 (Recurrent frozen geometry dichotomy)**.** *A recurrent frozen fixed-stage geometry cannot generate a genuinely new independent obstruction. It must fall into one of two classes:*

(a) almost-neutral periodic compatibility, or

(b) structural degeneration requiring deeper re-extraction.

[Status: Strong sketchsketch]

Proof sketch. On a recurrent frozen subsequence the combinatorial skeleton is constant. Therefore the normalized defect either returns with nonzero asymptotic size, which is periodic compatibility, or tends to zero, which is degeneration and re-extraction. A third independent possibility is excluded because no new local combinatorics is produced along the recurrent skeleton. □

6.3. Fixed-stage trichotomy

Proposition 6.5 (Fixed-stage classification)**.** *Every sufficiently deep fixed-stage block belongs to one of the following classes:*

(i) shrinking separation block, hence eventually frozen,

(ii) almost-neutral periodic block,

(iii) structural degeneration block.

[Status: Strong sketchsketch]

Proof sketch. If the block is in the separation case and the width shrinks, theorems 6.1 and 6.2 give eventual freezing. If the width does not shrink, apply theorems 6.3 and 6.4. Non-separation is treated as degeneration by definition of the branch decomposition. □

7. Global supply of independent blocks and rank depletion

The global argument no longer tracks individual same-stage candidates. Instead it tracks *genuinely new shrinking frozen blocks* modulo periodic and degenerate reuse.

Definition 7.1 (Independent shrinking frozen block)**.** A fixed-stage block is called an independent shrinking frozen block if it is

(i) in the separation case,

(ii) in the shrinking-width regime,

(iii) eventually frozen,

(iv) not periodic,

(v) not a degenerate re-extraction of a previously recorded obstruction.

Definition 7.2 (Rank of a new block)**.** To each independent shrinking frozen block B we assign

$$\mathcal{R}(B) = (\mathrm{edd}(B), \mathrm{do}(B), \mathrm{sci}(B)).$$

Here

- $\mathrm{edd}(B)$ is the first effective disagreement depth of the block-producing comparison;
- $\mathrm{do}(B)$ is the number of re-extractions still needed before the principal defect becomes nondegenerate at the current scale;
- $\mathrm{sci}(B)$ is a finite stage-complexity index, for instance

  $$\mathrm{sci}(B) = \#\mathcal{Q}^{\mathrm{adm}}(B) + \#\mathcal{M}(B) + \mathrm{tr}(B),$$

  where $\mathcal{Q}^{\mathrm{adm}}(B)$ is the admissible candidate set at that stage, $\mathcal{M}(B)$ is the set of active transition monitors, and $\mathrm{tr}(B)$ is the remaining transient length before complete freezing.

The order on ranks is lexicographic, with deeper effective disagreement interpreted as smaller in the first coordinate.

Lemma 7.3 (Rank decreases along genuinely new blocks)**.** *Let*

$$B_1, B_2, B_3, \dots$$

be genuinely new independent shrinking frozen blocks appearing along a single admissible branch. Then

$$\mathcal{R}(B_{j+1}) <_{\text{lex}} \mathcal{R}(B_j) \qquad \text{for all } j.$$

[Status: Strong sketchsketch]

Proof sketch. The first coordinate decreases because theorem 6.1 prevents reuse of the same effective same-stage ambiguity. If the first coordinate does not change, then a block with identical defect order is not genuinely new: it is periodic or degenerate. Thus the second coordinate must decrease. If both first and second coordinates remain the same, only local stage complexity differs; but freezing and transition rigidity imply that previously used ambiguity patterns do not reappear, so the third coordinate decreases. □

Corollary 7.4 (Well-founded depletion of new block supply)**.** *Along a single admissible branch, genuinely new independent shrinking frozen blocks can appear only finitely many times. [Status: Strong sketchsketch]*

Proof. The lexicographic order on $\mathbb{N}^3$ has no infinite descending chain. Apply theorem 7.3. □

8. Stage-transition rigidity and the absence of generic persistent local resonance

Theorem 8.1 (Stage-transition rigidity)**.** *Assume the local normal-form package and the fixed-stage classification above. Then any sufficiently deep admissible deterministic tail eventually enters one of the following regimes:*

(i) frozen regime,

(ii) almost-neutral periodic regime,

(iii) structural degeneration regime.

In particular, no independent persistent new obstruction branch survives indefinitely. [Status: Strong sketchsketch nearing final form]

Proof sketch. Every fixed-stage block is handled by theorem 6.5. Non-shrinking blocks become periodic or degenerate. Shrinking separation blocks eventually freeze and, by theorem 6.1, cannot regenerate the same ambiguity. By theorem 7.4, genuinely new shrinking frozen blocks occur only finitely often along a single branch. Hence after a finite depth no independent new obstruction remains; only frozen continuation, periodic recurrence, or degeneration survive. □

Theorem 8.2 (Absence of generic persistent local resonance)**.** *Assume:*

(a) block-affine exactness and local resonance reduction;

(b) candidate restriction and the local normal-form package of section 9;

(c) same-stage sharp counting from theorem 4.3;

(d) width control and deterministic-tail reduction;

(e) fixed-stage classification together with rank depletion.

Then the block-compressed local-resonance analysis admits no generic persistent local-resonance obstruction branch. *[Status: Near-final skeleton, not claimed as final proof]*

Proof. By theorem 5.2, sufficiently deep admissible branches have deterministic tails once width is dominated by spacing. By theorem 8.1, every deep deterministic tail eventually belongs to the frozen, periodic, or degenerate regime. The periodic and degenerate regimes do not produce genuinely new generic obstructions. The frozen regime cannot produce infinitely many genuinely new shrinking frozen blocks by theorem 7.4. Therefore no generic persistent local-resonance branch survives. □

9. The local normal-form package

This section isolates the final local analytic mechanism needed to make the sharp counting and one-sided admissibility fully rigorous.

9.1. Finite cell decomposition

Lemma 9.1 (Same-stage local cell decomposition)**.** *Fix a stage n and a comparison orientation. The same-stage relevant parameter set can be partitioned into finitely many cells*

$$\mathcal{T}_n^{\mathrm{rel}} = \mathcal{C}_1 \sqcup \cdots \sqcup \mathcal{C}_M$$

such that on each cell the following are fixed:

(i) compared local block skeleton,

(ii) first effective disagreement location,

(iii) local valuation/carry pattern,

(iv) principal sign pattern,

(v) the algebraic-affine recipe producing $(\Delta\lambda_t, \Delta\tau_t)$.

[Status: Strong sketchsketch]

Proof sketch. There are only finitely many ways for the discrete comparison data to change at a fixed stage: the local skeleton, disagreement location, valuation pattern, and admissibility sign pattern all come from finite choices. Group same-stage relevant parameters with identical discrete data into one cell. □

9.2. Cellwise reduced-defect linearization

Lemma 9.2 (Cellwise reduced-defect linearization)**.** *Fix one cell $\mathcal{C}$ from theorem 9.1. Then there exists a local parameter ξ_t with $\xi_t \asymp \delta_t$ on the cell and a fixed template map*

$$\mathcal{F}_{\mathcal{C}}(\xi) = (\Lambda_{\mathcal{C}}(\xi), B_{\mathcal{C}}(\xi))$$

such that

$$(\Delta\lambda_t, \Delta\tau_t) = \mathcal{F}_{\mathcal{C}}(\xi_t) - \mathcal{F}_{\mathcal{C}}(0).$$

If $\mathcal{F}_{\mathcal{C}}$ is differentiable at $\xi = 0$, then

$$(\Delta\lambda_t, \Delta\tau_t) = \delta_t(A_{\mathcal{C}}, B_{\mathcal{C}}) + o(\delta_t).$$

If $(A_{\mathcal{C}}, B_{\mathcal{C}}) = (0, 0)$, then the cell is classified as degenerate. *[Status: Needs full proof: final local analytic derivation from explicit formulas]*

Proof sketch. Inside a fixed cell the discrete template is frozen, so the comparison coefficients are given by one algebraic-affine recipe. The same-stage parameter enters this recipe through a single local deviation variable. Differentiability of that recipe at the cell origin then yields the first-order expansion. Vanishing first-order coefficient means the principal defect collapses at that scale, hence degeneration. □

Corollary 9.3 (Orientation-compatible window selection)**.** *Assume theorem 9.2. Then on every nondegenerate cell the principal sign pattern is determined by* $\operatorname{sgn}(\delta_t)$ *for sufficiently deep relevant parameters. Consequently the admissible window accepts at most one sign of* δ_t*, hence at most one side of the center* u_**.* *[Status: Proof sketch, one step from full rigor]*

Proof. The first-order term determines the principal sign pattern, and a fixed comparison orientation accepts only one such pattern. □

10. What remains to turn the skeleton into a final proof

The manuscript is close to a full proof-oriented paper, but two final formalization tasks remain.

Task 1. Explicit local cell formulas. For each fixed cell from theorem 9.1, write the local affine comparison recipe in explicit form and prove differentiability of the template map required by theorem 9.2.

Task 2. Final local one-sidedness closure. Once theorem 9.2 is formal, upgrade theorem 4.2 and theorem 4.3 from strong sketches to proved theorems.

Everything else in the present manuscript is already organized to absorb those two inputs: the fixed-stage classification, recurrence dichotomy, rank depletion, and stage-transition rigidity all become theorem-level once the cellwise normal form is written from the actual local formulas.

11. Conclusion

This paper stops just short of a formal complete-proof claim, but it now presents the local-resonance route in a form that is very close to completion. The main point is no longer merely that a promising skeleton exists; rather, the global architecture has essentially stabilized, and the genuine remaining burden has been pushed into a sharply localized local normal-form package.

At the present stage, the broad architecture is as follows:

- local resonance is reduced to required-height matching at first effective disagreement;
- same-stage candidates are sparse and, modulo one-sided admissibility, sharply bounded;
- deep branches become deterministic;
- fixed-stage deterministic behavior falls into frozen, periodic, or degenerate regimes;
- genuinely new shrinking frozen blocks are depleted by a well-founded rank.

Thus the last unresolved issue is local and analytic, not architectural: the cellwise linearization of the reduced defect map. Once that package is completed from explicit formulas, the present draft is intended to upgrade naturally from a status-aware working paper to a final theorem paper.

A. Notation index

Symbol	Meaning
$a = (N, T)$	One local-minimum block
$F_a(m) = \lambda(a)m + \tau(a)$	Odd-to-odd block map
$\Lambda(w), B(w)$	Multiplicative and additive coefficients of a word w
Ξ^{req}	Required height for exact coincidence at first effective disagreement
$\theta_n = \|q_n\alpha\|$	Principal Diophantine error at stage n
$m_t = q_{n-1} + tq_n$	Semiconvergent family
$u_* = \theta_{n-1}/\theta_n$	Same-stage center
$\delta_t = \theta_{n-1} - t\theta_n$	Oriented same-stage deviation
I_d^{res}	Admissible enclosure of required heights at depth d
$\mathrm{wd}(I_d^{\mathrm{res}})$	Width of the admissible enclosure
$\mathrm{edd}, \mathrm{do}, \mathrm{sci}$	Rank coordinates for new shrinking frozen blocks

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